IMPACT TANK ’26 · ROUND 1 · PROBLEM DISCOVERY BRIEF

The Journey After Harvest

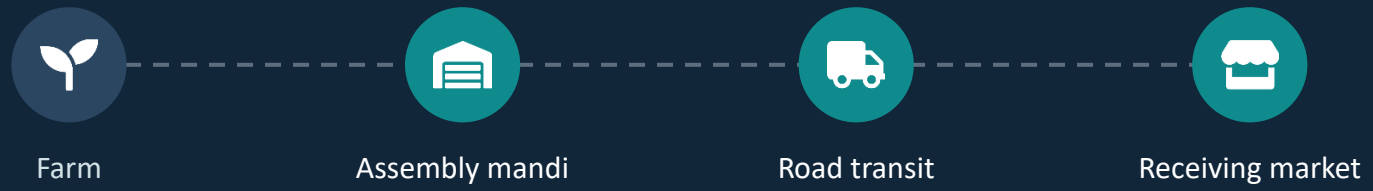
Why perishable fruit and vegetables lose quality and value between the mandi and the receiving market in India — and why that loss is still poorly understood

19.3 Mt

million tonnes (Mt) of fruit and vegetables lost after harvest in India each year, across all stages from farm to retail

NABCONS for MoFPI, reference period 2020–22 (fruits 7.36 Mt + vegetables 11.97 Mt) [1]

This brief focuses on the stretch from aggregation to sale



Team: [Team name]
Members: [Member 1] · [Member 2] · [Member 3] · [Member 4] · [Member 5]
Institution: [College / University]

Evidence key DOCUMENTED | INTERPRETATION | HYPOTHESIS · VALIDATE

Perishable loads lose value between dispatch and sale

PROBLEM STATEMENT

TARGET USER INTERPRETATION	Dispatchers of perishable fruit and vegetable consignments: commission agents, traders and FPO marketing staff who send produce from assembly mandis to distant wholesale markets
FACE DIFFICULTY WITH DOCUMENTED	getting each load to a market where it arrives still saleable and fetches a fair price
BECAUSE OF HYPOTHESIS · VALIDATE	transit time, heat, handling and destination-market conditions that vary after dispatch, in mostly non-refrigerated trucks, and that are poorly measured or visible when the decision is made
RESULTING IN DOCUMENTED	quality downgrades, distress sales and wastage, with costs falling largely on farmers and traders and reaching consumers as price volatility

IN SCOPE

- Highly perishable fruit & vegetables (e.g. tomato, guava, banana, mango, gourds)
- Road movement after first aggregation, up to sale at the receiving market
- Ambient (non-reefer) transport, the dominant mode [4]
- Variation by distance, season and region

OUT OF SCOPE

- Grains & pulses: market-stage loss only ≈0.6–1.2% [1]
- Farm-level harvest losses; household waste
- Long-term cold storage of potato and onion
- Any proposed solution

Tags show how firmly each part is supported. The “because” clause mixes documented transit factors [3] with an untested claim about visibility at dispatch. Sources: [1] NABCONS 2022 via PIB 2025; [3] WRI India 2024; [4] NCCD 2015.

Evidence key DOCUMENTED | INTERPRETATION | HYPOTHESIS · VALIDATE

Where it happens: failure points from mandi to market

FOCUS OF THIS BRIEF

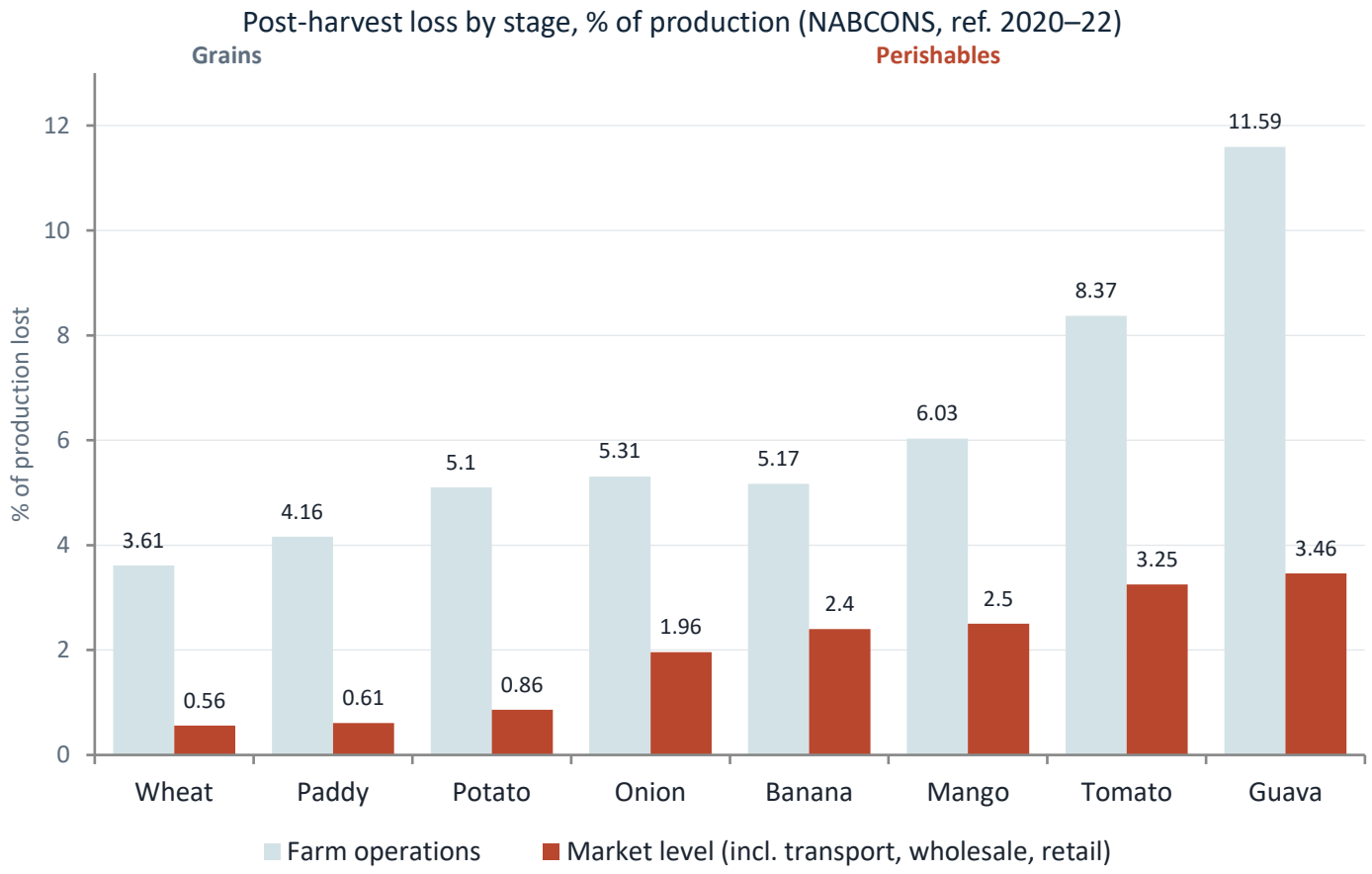


Farm & harvest	Assembly mandi *aggregation, sale to trader*	Packing & dispatch *loading, choice of market*	Road transit *hours to days*	Market entry *waiting, unloading*	Wholesale → retail *sale, resale*
Harvest maturity is chosen to suit travel distance [3] DOCUMENTED	Poor sorting and grading at origin show up later as transit loss [3] DOCUMENTED	Crates vs polythene bags; crates overfilled when paid per crate [3] DOCUMENTED	Heat and vibration on poor roads damage produce [3] DOCUMENTED	Waiting outside the market; trade mostly in morning hours [3] DOCUMENTED	A glut on arrival lowers price; damaged lots sell for less [3] DOCUMENTED
Farm stage carries the largest share of tomato loss: 8.37% [1] DOCUMENTED	Destination picked without sight of arrivals at other markets HYPOTHESIS · VALIDATE	Non-refrigerated trucks; none used in the study area [3] DOCUMENTED	Delays from inter-state permits, traffic and breakdowns [3] DOCUMENTED	No storage at the market, so unsold lots go at distress prices [3] DOCUMENTED	Delhi receives tomato all year from many states; long hauls are routine [5] DOCUMENTED

HYPOTHESIS · VALIDATE *The field evidence comes from short, local flows (markets 25–30 km away, Madhya Pradesh tomato) [3]. Whether these failure points grow on long inter-state hauls has not been tested.*

Sources: [1] NABCONS 2022, Annexure-II via PIB (1 Aug 2025); [3] Lamba, Singh & Anand, WRI India (2024), Tomato trail, Madhya Pradesh field study; [5] Roy et al., RBI WPS 08/2024.

Fruit and vegetables lose 3–6× more than grains at the market stage



19.3 Mt

fruit (7.36 Mt) + vegetables (11.97 Mt) lost a year, all stages farm to retail [1]

₹1.53 lakh crore

annual value of post-harvest loss across all 54 commodities studied, not F&V alone (2020–21 production, 3-yr avg prices) [1]

INTERPRETATION

Read with care: “market level” bundles transport, wholesale, retail and processing. No transport-only share is published here, so these figures bound the problem; they do not measure transit loss.

[1] NABCONS (2022) for MoFPI, as tabled in Rajya Sabha reply, PIB release 2151371 (1 Aug 2025), Annexure-II Tables 1–2. Earlier ICAR-CIPHET (2015) estimate [2]: ₹92,651 crore at 2014 prices. Methods and base years differ, so the two are not a trend.

The dispatcher decides the journey; the farmer often pays for it

PRIMARY USER

The dispatcher

Commission agent, trader or FPO marketing lead at the assembly mandi

Chooses the destination, timing and packing for a load they cannot observe once it leaves. Field evidence shows these wholesale-level choices of packaging and destination distance shape shelf life [3].

Why primary: the difficulty starts at this decision point, and it is narrow enough to study directly.

SECONDARY STAKEHOLDERS: affected by the outcome

Farmer / FPO member

Damaged lots sell for less, and that loss is borne mostly by the farmer [3]. Farmers get about 33% of the Delhi consumer price for tomato [5].

Transporter / driver

Deals with heat, rough roads, permit checks, waiting at market gates and breakdowns, in trucks without refrigeration [3].

Receiving wholesaler / retailer

No storage on site, so stock must sell the same day. In the study, 12% of tomatoes were wasted at retail and 11% sold as Grade II at about half price [3].

Consumer

Perishability and short crop cycles make tomato, onion and potato prices swing sharply [5].

SHAPE THE CONDITIONS

- APMC boards & state governments
- e-NAM / Ministry of Agriculture
- MoFPI & NCCD (cold chain)
- Permits, roads & checkpoints

Stakeholder difficulties come from published studies; we have not yet run our own interviews. [3] WRI India 2024 (MP tomato chain); [5] RBI WPS 08/2024 (Delhi value chain).

Evidence key DOCUMENTED | INTERPRETATION | HYPOTHESIS · VALIDATE

Why it persists: produce moves warm through an unmeasured chain

UNDERLYING CONDITIONS	PROXIMATE CAUSES	OBSERVABLE SYMPTOMS
Cold chain built for storage, not movement: an 85% shortfall in reefer vehicles and 99.6% in pack-houses (2014–15) [4] DOCUMENTED	Heat exposure: perishables deteriorate 2–3× faster for every 10°C rise [6] DOCUMENTED	Produce downgraded on arrival (Grade II sold at about half price) [3] DOCUMENTED
A fragmented chain with many intermediaries; decisions are split across actors [5] DOCUMENTED	Delay: inter-state permits, waiting outside markets, traffic, breakdowns [3] DOCUMENTED	Distress and same-day selling at markets and retail [3] DOCUMENTED
Transit conditions and outcomes are not recorded in any public dataset (slide 9) DOCUMENTED	Mechanical damage: vibration, poor roads, polythene bags, overfilled crates [3] DOCUMENTED	Physical wastage at the market stage: 2–3.5% for most F&V [1] DOCUMENTED
Dispatchers lack a timely view of conditions at destination and on the way HYPOTHESIS · VALIDATE	Arrival mismatch: produce reaches a glutted market with no storage to wait in [3][5] DOCUMENTED	Sharp price swings across seasons and markets [5] DOCUMENTED

How much each cause contributes in transit has not been quantified in published Indian data. Consequences are on slide 8. Sources: [1] NABCONS 2022; [3] WRI India 2024; [4] NCCD 2015; [5] RBI 2024; [6] Kader & Rolle, FAO AGS Bulletin 152 (2004).

Today's efforts address storage, prices and trucks more than the trip

Approach	What it does	Addresses	Gap that remains within our scope	Evidence
Cold-chain infrastructure (PMKSY Cold Chain scheme; NCCD planning)	Funds cold stores, pack-houses, reefer transport	Heat exposure, holding	Shortfall concentrated in movement: reefers 85%, pack-houses 99.6% (2014–15); storage skewed to bulk, single-commodity stores such as potato	**Documented [4]**
Digital market information & trading (Agmarknet; e-NAM, 1,656 mandis by Mar 2026)	Daily arrivals and prices; online bidding across mandis	Price discovery, buyer reach	Records what arrived and at what price, not trip time or arrival condition. Government calls logistics “a major hurdle” for inter-mandi trade	**Documented [7][8]; dataset scope: interpretation [9]**
Kisan Rath app (MoA&FW, 2020)	Matches consignors with trucks and aggregators	Truck availability	Downloads fell over time; over half of users rely on local transport outside the app	**Documented [10]**
Operation Greens “TOP to TOTAL”: 50% freight subsidy on Kisan Rail	Subsidised moving notified F&V from surplus to consumption areas	Cost of long hauls	Time-bound pilot (Oct 2020–Mar 2021), rail-only; current status not verified	**Documented [11]**
Navigation & route-optimisation tools (e.g. Google Maps, Route Optimization API)	Traffic-aware ETAs; time windows, vehicle capacity, cost per km/hour	Travel time, fleet planning	No built-in model of produce quality decay in public documentation; shelf life can only be approximated as a time window	**Interpretation of public docs [12]; validate**
Research on perishable-goods routing	Models quality decay inside vehicle-routing problems	Planning under perishability	Review of 65 papers: dynamic, quality-aware routing still open; no agreed decay model; mostly cold-chain settings	**Documented [13]; India-ambient gap: to confirm**

None of these is ineffective. Each targets a different part of the problem. [4] NCCD 2015; [7] Rajya Sabha reply, 7 Feb 2025; [8] PIB, Apr 2026; [9] agmarknet.gov.in; [10] Shalendra et al. 2021 (MANAGE); [11] MoFPI / Lok Sabha reply on Operation Greens; [12] Google Route Optimization API docs; [13] Awad, Ndiaye & Osman 2021.

Costs fall on farm income, market quality and consumer prices

FARMER

~33%

farmers' share of the Delhi consumer price for tomato (onion ~36%, potato ~37%). The rest covers transport, fees, losses and margins [5]

TRADER

5–10%

weight loss traders absorb on onion: about 5% for the rabi crop, about 10% for kharif [5]

RECEIVING MARKET

12%

of tomatoes wasted at retail by day's end in the MP study, with another 11% sold as Grade II at about half price [3]

CONSUMER

Volatile

TOP prices swing sharply, driven partly by perishability and short seasonal cycles, and feed into food and headline inflation [5]

Food system: 19.3 Mt of F&V lost each year across all stages [1]. In the MP study districts, tomato lost at wholesale and retail adds about 6,824 t CO_2e a year from landfill [3].

INTERPRETATION

These costs build up across the whole chain. With current data, the share that starts in transit cannot be separated out. That is a gap in itself.

[1] NABCONS 2022 via PIB 2025; [3] WRI India 2024, Madhya Pradesh tomato chain (direct measurement, 2020–21 production base); [5] Roy et al., RBI WPS (DEPR) 08/2024, value-chain estimates for Delhi.

Three distinct gaps: in data, in research and in everyday practice

DATA GAP · verified

- Fresh F&V need no GST e-way bill (CGST Rule 138(14)(e)), so India's main digital freight record largely misses these trips [14]
- Agmarknet and e-NAM record arrivals, prices and trades, not trip duration, route or condition on arrival [8][9]
- We found no public trip-level spoilage or rerouting-decision dataset [15]
- Indian loss studies rely more on perception surveys than on direct measurement [3]

RESEARCH GAP · partly established

- Dynamic, quality-aware routing is still an open research area, with no agreed decay model [13]
- Most quality-decay work assumes refrigerated settings; India's ambient mandi-to-market flows look thinly studied (to confirm by systematic review)
- Loss studies split losses into farm and market stages, not transit vs waiting vs handling [1]

PRACTICE GAP · hypothesis

- How dispatchers choose destination and timing today, and what information they use: we found no published evidence
- Whether problems en route (delay, heat, a glut at destination) are noticed and acted on after dispatch
- Which signals (prices elsewhere, weather, road delays) dispatchers can actually access when they decide

OPEN QUESTIONS THIS BRIEF RAISES

1 What share of market-level loss arises in transit, in waiting, and in handling?

2 How does it vary with distance, season (summer heat, monsoon) and region?

3 How often do loads arrive into a glut, and could that have been known at dispatch?

4 Which crops are most sensitive to delay, and which to heat?

Scarce data shows the problem is under-measured, not that it is large. Its size in transit has to be established, not assumed.

[1] NABCONS 2022; [3] WRI India 2024; [8] PIB 2026; [9] agmarknet.gov.in; [13] Awad et al. 2021; [14] CGST Rules 2017, r.138(14)(e) with Notif. 2/2017-CT(Rate) Schedule (fresh fruits & vegetables exempt); [15] team research blueprint, data-availability audit (internal).

Why this problem deserves investigation now

It is meaningful

F&V lose 19.3 Mt a year, and three to six times more than grains at the market stage [1]; farmers keep about a third of the consumer price [5].

Many gain from understanding it

Farmers and FPOs who absorb downgrades, the dispatchers who choose markets, receiving traders, and agencies planning cold-chain and market investment [4].

The gaps are real

Transit outcomes are not in public data, quality-aware dynamic routing is an open research question, and we found no published study of how dispatchers decide (slide 9).

It is specific and researchable

Bounded by crop (highly perishable F&V), leg (mandi to receiving market) and mode (ambient road), and testable with public market data plus field measurement.

THE PROBLEM IN ONE SENTENCE

In India, perishable fruit and vegetables lose value on the road between mandi and market: a stretch that is rarely refrigerated, rarely measured, and little studied from the point of view of the people who dispatch the load.

[1] NABCONS 2022 via PIB 2025; [4] NCCD 2015; [5] RBI WPS 08/2024. Our next discovery step is to check the primary-user hypothesis directly with dispatchers.

Sources, numbered as cited on each slide (1/2)

		USED ON
[1]	NABARD Consultancy Services (NABCONS) for Ministry of Food Processing Industries (2022). Study to Determine Post-Harvest Losses of Agri Produces in India (ref. period 2020–22). Figures as tabled in Rajya Sabha, PIB release ID 2151371, 1 Aug 2025, Annexures I–II. https://www.pib.gov.in/PressReleasePage.aspx?PRID=2151371	*Slides 1, 3, 4, 6, 8, 9, 10*
[2]	Jha, S.N. et al. / ICAR-CIPHET (2015). Assessment of Quantitative Harvest and Post-Harvest Losses of Major Crops and Commodities in India. Total loss value ₹92,651 crore, as reported in a Lok Sabha reply. https://eparlib.nic.in/bitstream/123456789/706362/1/56899.pdf	*Slide 4 (footnote)*
[3]	Lamba, S., Singh, R. & Anand, S. (2024). Tomato Trail: Tracking Food Loss and Food Waste in Madhya Pradesh. Working Paper, WRI India, Sept 2024. https://wri-india.org/sites/default/files/Tomato-trail-Tracking-food-loss-food-waste-Madhya-Pradesh.pdf	*Slides 3, 5, 6, 8, 9*
[4]	National Centre for Cold-chain Development & NABCONS (2015). All India Cold-chain Infrastructure Capacity (Assessment of Status & Gap). Requirement vs created, 2014–15. https://nccd.gov.in/uploads/All_India_Cold_Chain_Infrastructure_2015_5da1279baa.pdf	*Slides 2, 6, 7, 10*
[5]	Roy, R., Gupta, S., Wardhan, H., Sarkar, S., Tewari, S., Bansal, R., Bhatia, S. & Gulati, A. (2024). Vegetables Inflation in India: A Study of Tomato, Onion and Potato (TOP). RBI Working Paper Series (DEPR) No. 08/2024. https://rbi.org.in/Scripts/PublicationsView.aspx?id=22723	*Slides 3, 5, 6, 8, 10*
[6]	Kader, A.A. & Rolle, R.S. (2004). The Role of Post-Harvest Management in Assuring the Quality and Safety of Horticultural Produce. FAO Agricultural Services Bulletin 152, Section 3.2. https://www.fao.org/4/y5431e/y5431e04.htm	*Slide 6*
[7]	Rajya Sabha (2025). Ministry of Agriculture & Farmers Welfare, reply to Question 267 on e-NAM, 7 Feb 2025 (inter-state trade; logistics as a hurdle; e-NAM 2.0). https://rsdebate.nic.in/bitstream/123456789/758227/1/PQ_267_07022025_S46_p1_p12.pdf	*Slide 7*
[8]	Press Information Bureau (2026). e-NAM backgrounder, April 2026: 1,656 mandis in 23 states and 4 UTs onboarded. https://static.pib.gov.in/WriteReadData/specificdocs/documents/2026/apr/doc2026413845801.pdf	*Slides 7, 9*

All URLs were checked in October 2026. Parliamentary figures are as tabled on the date given. No stakeholder interviews, field visits or surveys of our own are reported in this deck.

Sources, numbered as cited on each slide (2/2)

		USED ON
[9]	Directorate of Marketing & Inspection, MoA&FW. AGMARKNET portal: daily commodity arrivals and prices reported by markets. https://agmarknet.gov.in	*Slides 7, 9*
[10]	Shalendra, Kumar, R., Gummagolmath, K.C. & Angadi, S. (2021). Transportation in Agriculture: A Case of Kisan Rath App. Journal of Agricultural Extension Management (MANAGE). https://epubs.icar.org.in/index.php/JAEM/article/download/159460/62590/489185	*Slide 7*
[11]	Ministry of Food Processing Industries (2020–21). Operation Greens "TOP to TOTAL": 50% freight subsidy for notified fruits & vegetables on Kisan Rail. Lok Sabha Unstarred Q. 2423. https://eparlib.nic.in/bitstream/123456789/992164/1/AU2423.pdf	*Slide 7*
[12]	Google Maps Platform (2025). Route Optimization API: parameter list and ShipmentModel reference (time windows, load demands, cost per hour/km). https://developers.google.com/maps/documentation/route-optimization/parameter-list	*Slide 7*
[13]	Awad, M., Ndiaye, M. & Osman, A. (2021). Vehicle routing in cold food supply chain logistics: a literature review. The International Journal of Logistics Management, 32(2), 592–617. https://doi.org/10.1108/IJLM-02-2020-0092	*Slides 7, 9*
[14]	Central Goods and Services Tax Rules, 2017, Rule 138(14)(e) (as substituted by Notification 12/2018-Central Tax) read with the Schedule to Notification 2/2017-Central Tax (Rate). https://www.gstzen.in/a/rule-138-central-goods-and-services-tax-rules-2017-cgst-rule-138-replaced-as-per-central-tax-notification-122018.html	*Slide 9*
[15]	[Team name] (2026). Agricultural Supply Chain Research Blueprint: dataset inventory and data-availability classification (internal document). *Internal, not published*	*Slide 9*

[14] is our reading of the e-way bill rule; check it against the current CBIC text before relying on it operationally. [15] is our own planning document, cited for its data-availability audit, not as external evidence.